

Name _____ Date _____

10.1 Logic expressions

Getting started

Complete the truth tables for the following gates:

1 AND

X	Y	X AND Y
0	0	
0	1	
1	0	
1	1	

2 OR

X	Y	X OR Y
0	0	
0	1	
1	0	
1	1	

3 NOT

X	NOT X
0	
1	

4 NAND

X	Y	X NAND Y
0	0	
0	1	
1	0	
1	1	

5 NOR

X	Y	X NOR Y
0	0	
0	1	
1	0	
1	1	

6 XOR

X	Y	X XOR Y
0	0	
0	1	
1	0	
1	1	

Practice

1 Evaluate each statement to identify if the output is 1 (true) or 0 (false).

- a 1 AND 0
- b 1 OR 1
- c 1 NOR 0
- d 1 XOR 1

e NOT(1)

f 1 NAND 0

g 1 AND 1

h 0 OR 0

i 0 NOR 0

j 1 XOR 0

2 Complete the truth table for the following logic expressions.

a

A	B	C	A AND B AND C
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

b

A	B	C	A XOR (B OR C)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

c

A	B	C	(A AND B) OR (C XOR B)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

d

A	B	C	(NOT A NAND B) NOR C
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

e

A	B	C	(C XOR A) AND (B OR A)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Challenge

Write a logic expression that makes use of **four** inputs; A, B, C and D.

Create the truth table for your logic expression.

Swap the expression with another learner and draw the truth table for their logic expression.

Compare your truth tables and see if they match.